UbuntuNet Alliance Women Hackathon 2026

Phase I - Week 2 Submission

SOLARTRACE

Chapter 2: Technical Architecture & Design

Team South Sudan

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Technical Specifications, Component Design, Interface Design

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1 INTRODUCTION TO SYSTEM DESIGN

This chapter presents the comprehensive technical architecture and design of the SolarTrace system. Following the foundation established in Chapter 1, this document translates the problem statement and objectives into a detailed technical blueprint. The design phase is critical for ensuring that the prototype development proceeds efficiently and that all components work together harmoniously.

The chapter is organized as follows:

- **Alternative Systems Analysis**: Examination of existing solutions
- **System Architecture**: High-level and detailed component interactions
- **Context Diagram**: External entity interactions
- **Data Flow Diagrams**: Information flow through the system
- **Use Case Diagram**: User-system interactions
- **Activity Diagrams**: Process workflows
- **Sequence Diagrams**: Object interactions over time
- **Class Diagrams**: Logical code structure
- **Package Diagrams**: Module organization
- **Technical Specifications**: Component-level details

2 ALTERNATIVE SYSTEMS ANALYSIS

2.1 Overview

Before finalizing the SolarTrace design, an analysis of existing solar tracking systems was conducted to identify strengths, weaknesses, and opportunities for innovation.

2.2 Existing System 1: Abid (2017) - Arduino Based Blind Solar Tracking Controller

Table 1: Analysis: Abid (2017) Solar Tracking System

Aspect	Details
Technology	Arduino-based blind tracking (no light sensors)
Tracking Method	Time-based algorithm using RTC
Target Application	General solar panel tracking
Strengths	Low cost, no sensors required, predictable operation
Weaknesses	Not designed for cooking applications, no temperature control, no IoT
Relevance to SolarTrace	Validation of RTC-based approach; SolarTrace extends with cooking modes, IoT, and sterilization

2.3 Existing System 2: Mahmood et al. (2025) - Hybrid Solar Cooker

Table 2: Analysis: Mahmood et al. (2025) Hybrid Solar Cooker

Aspect	Details
Technology	Hybrid solar cooker with tracking
Tracking Method	Light sensor-based automatic tracking
Target Application	Cooking in Iraq
Strengths	Demonstrated temperature improvement with tracking (95°C vs 86°C), hybrid design
Weaknesses	No IoT integration, no mobile application, no sterilization mode
Relevance to SolarTrace	Confirms tracking benefits; SolarTrace adds IoT, mobile app, and sterilization

2.4 Existing System 3: Rashid et al. (2025) - Hybrid Solar Geyser-Stove System

Table 3: Analysis: Rashid et al. (2025) Hybrid Solar System

Aspect	Details
Technology	IoT-enabled solar geyser-stove system
Target Application	Water heating and cooking
Strengths	IoT integration, real-time monitoring
Weaknesses	No automatic tracking, limited to specific geography
Relevance to SolarTrace	Validates IoT approach; SolarTrace adds automatic tracking and sterilization

2.5 Key Findings and SolarTrace Positioning

The analysis reveals that while individual components of SolarTrace exist in various systems, no single solution combines all the following features:

1. RTC-based automatic sun tracking
2. IoT integration with mobile application
3. Multiple cooking modes including sterilization
4. Offline data logging capability
5. Safety mechanisms for overheating protection
6. Design specifically for the South Sudan context

SolarTrace fills this gap by integrating all these features into a cohesive, low-cost system.

3 SYSTEM ARCHITECTURE

3.1 High-Level Architecture

The SolarTrace system follows a three-tier architecture:

1. **Physical Layer**: Hardware components including the parabolic reflector, servo motor, sensors, and microcontroller

2. **Control Layer**: Embedded firmware that processes sensor data, controls actuators, and manages communication

3. **Application Layer**: Blynk-based mobile application for user interaction, data visualization, and remote monitoring

3.2 ESP32-DEVKIT Microcontroller

The ESP32-DEVKIT serves as the main processing unit for the SolarTrace system. Figure 1 shows the ESP32-DEVKIT pinout diagram.

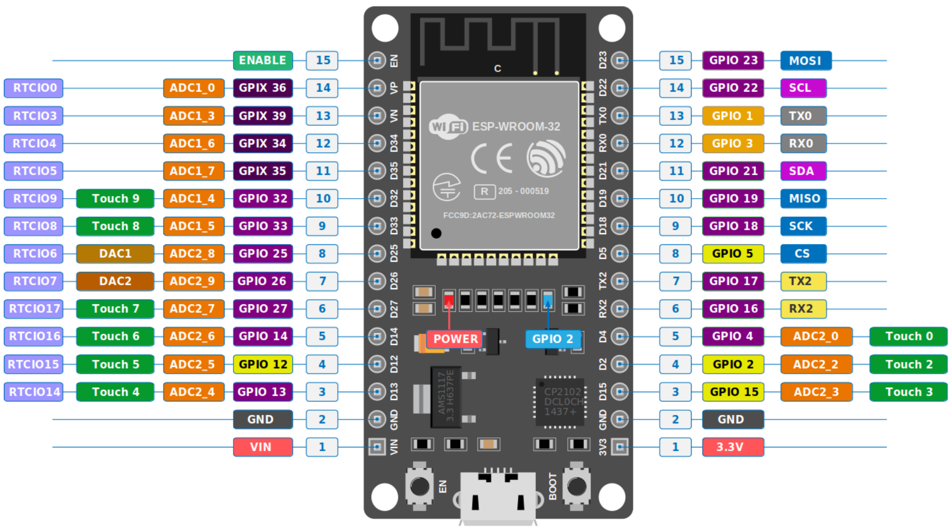

Figure 1: ESP32-DEVKIT Pinout Diagram

3.3 ESP32-DEVKIT Specifications

Table 4: ESP32-DEVKIT Technical Specifications

Specification	Value
Microcontroller	Tensilica Xtensa LX6 Dual-Core
Clock Speed	Up to 240 MHz
SRAM	520 KB
Flash Memory	4 MB (External)
WiFi	802.11 b/g/n (2.4 GHz)
Bluetooth	BLE 4.2
Operating Voltage	3.3V
Input Voltage	5V (via USB) / 7-12V (via VIN)
GPIO Pins	38 (25 usable)
ADC Channels	18 (12-bit)
DAC Channels	2 (8-bit)
USB-to-Serial	CP2102 or CH340
Board Dimensions	52mm x 28mm

3.4 System Architecture Diagram

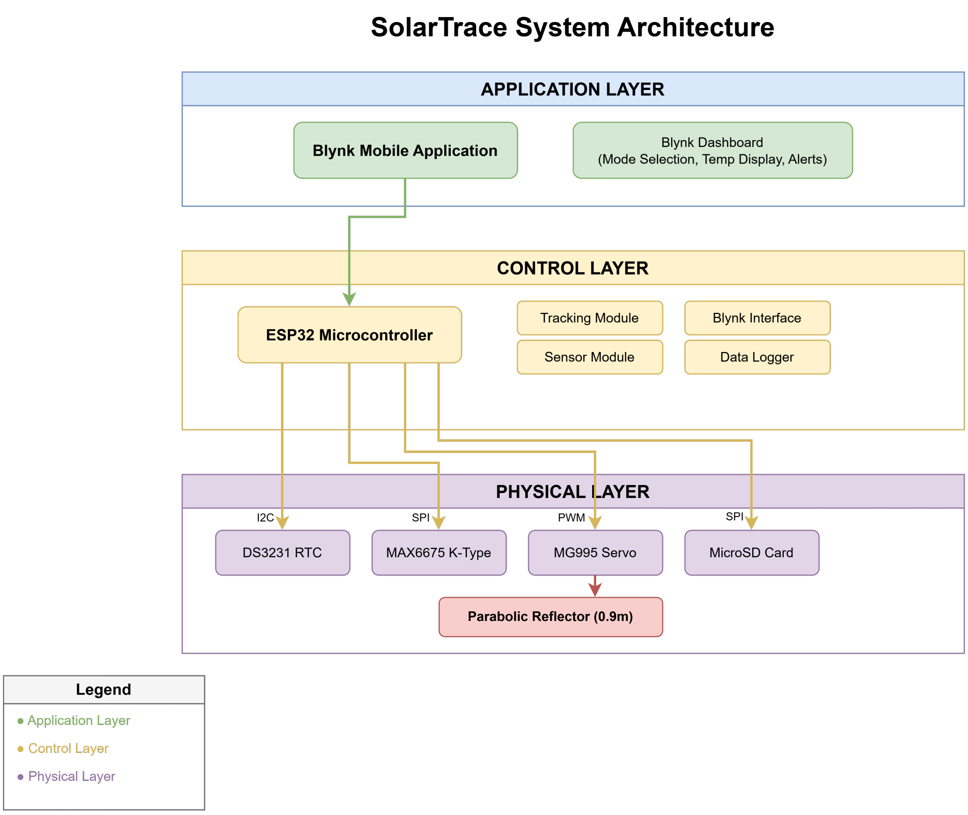

Figure 2: SolarTrace System Architecture Diagram

3.5 Component Descriptions

Table 5: System Component Description

Component	Model	Purpose
Microcontroller	ESP32-DEVKIT	Main processing unit; Wi-Fi/BLE connectivity; runs control logic
RTC Module	DS3231	Maintains accurate time; enables time-based sun tracking even when powered off
Temperature Sensor	MAX6675 + K-type	Measures cooking temperature up to 500°C; critical for mode control and safety
Servo Motor	MG995	Positions parabolic reflector; high torque (15kg/cm) for reliable movement
SD Card Module	MicroSD	Offline data logging; stores session data without internet connectivity
Power System	5V Panel + 18650	Provides off-grid power; battery backup for RTC and continuous operation

4 SYSTEM SCHEMATIC & WIRING DIAGRAM

4.1 Complete System Schematic

Figure 3 shows the complete electrical schematic of the SolarTrace system, including all components and their interconnections.

schematic_diagram.png

Figure 3: Complete System Schematic Diagram

4.2 Wiring Diagram

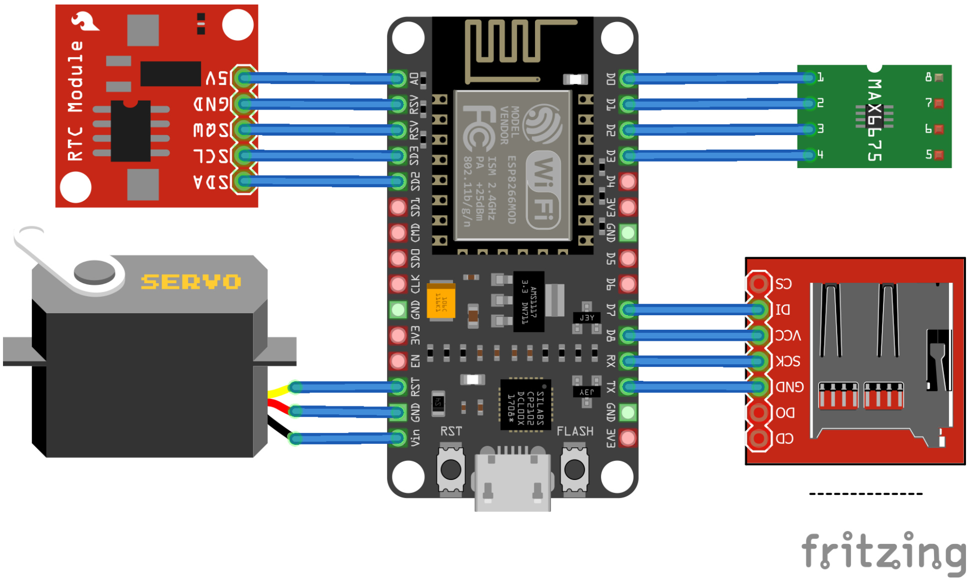

Figure 4: SolarTrace Wiring Diagram (Fritzing)

4.3 Detailed Pin Connection Table for ESP32-DEVKIT

Table 6: Complete Pin Connection Table for ESP32-DEVKIT

ESP32-DEVKIT Pin	Component	Component Pin	Function
3.3V	DS3231 RTC	VCC	Power (3.3V)
GND	DS3231 RTC	GND	Ground
GPIO21	DS3231 RTC	SDA	I2C Data
GPIO22	DS3231 RTC	SCL	I2C Clock
3.3V	MAX6675	VCC	Power (3.3V)
GND	MAX6675	GND	Ground
GPIO15	MAX6675	CS	Chip Select
GPIO14	MAX6675	SCK	SPI Clock
GPIO12	MAX6675	SO	Data Out
VIN (5V)	MG995 Servo	VCC	Power (5V)
GND	MG995 Servo	GND	Ground
GPIO13	MG995 Servo	Signal	PWM Control
3.3V	MicroSD	VCC	Power (3.3V)
GND	MicroSD	GND	Ground
GPIO5	MicroSD	CS	Chip Select
GPIO18	MicroSD	SCK	SPI Clock
GPIO23	MicroSD	MOSI	Master Out
GPIO19	MicroSD	MISO	Master In

4.4 I2C Pull-up Resistor Configuration

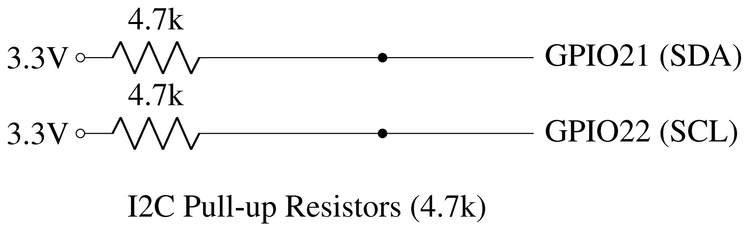

Figure 5: I2C Pull-up Resistor Configuration

5 CONTEXT DIAGRAM

The Context Diagram shows SolarTrace as a single process and its interactions with external entities.

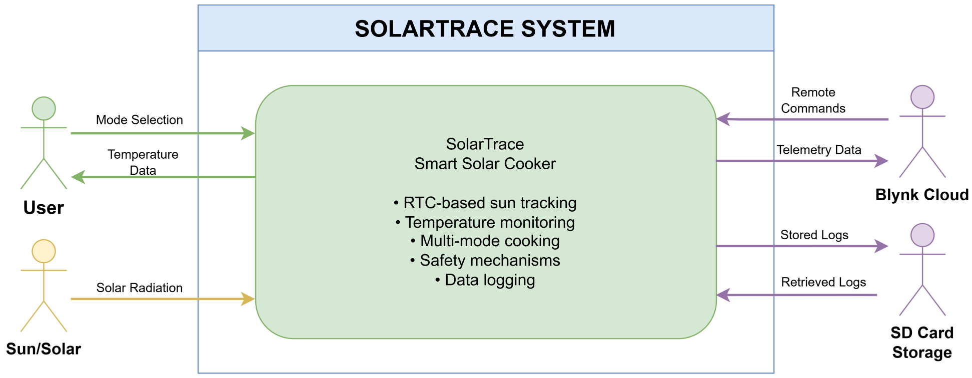

Figure 6: SolarTrace Context Diagram

External Entity Interactions:

- **User**: Selects cooking mode, receives temperature information and notifications
- **Sun**: Provides solar radiation for cooking and power
- **Blynk Cloud**: Facilitates remote monitoring and control
- **SD Card**: Stores session data for offline logging

6 DATA FLOW DIAGRAM

6.1 Level 0 DFD: Main System Functions

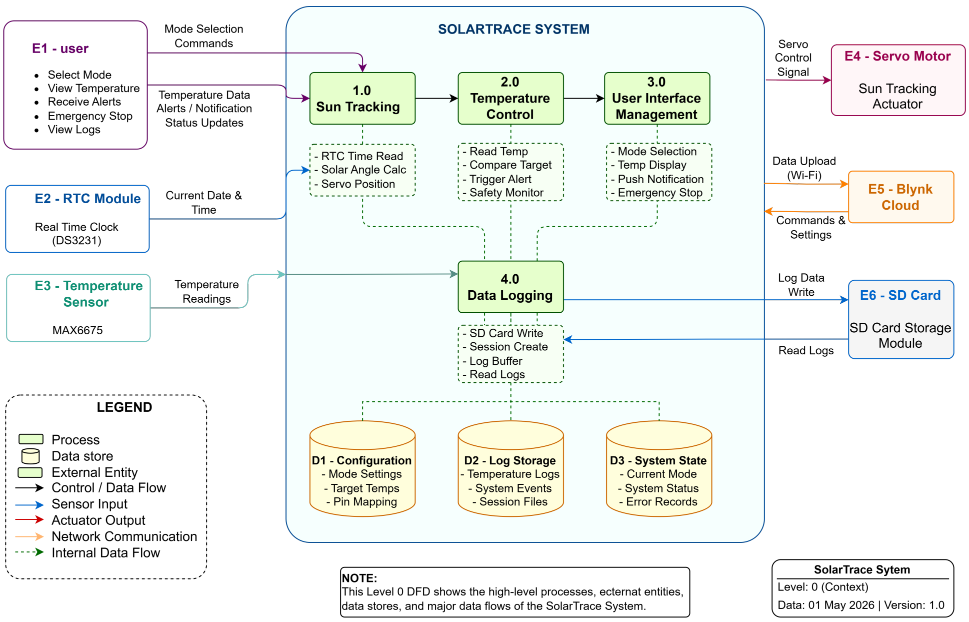

Figure 7: Level 0 Data Flow Diagram

6.2 Process Descriptions

Table 7: DFD Process Descriptions

Process	ID	Description
Sun Tracking	1.0	Reads RTC time, calculates solar angle, positions servo motor accordingly
Temperature Control	2.0	Reads thermocouple data, compares to target temperature, triggers alerts
User Interface Management	3.0	Handles mode selection, displays temperature, sends notifications via Blynk
Data Logging	4.0	Records session data to SD card for offline analysis

7 USE CASE DIAGRAM

7.1 System Actors

- **Primary Actor**: User (rural household, clinic staff)
- **Secondary Actors**: Blynk Cloud, SD Card
- **System**: SolarTrace

7.2 Use Case Diagram

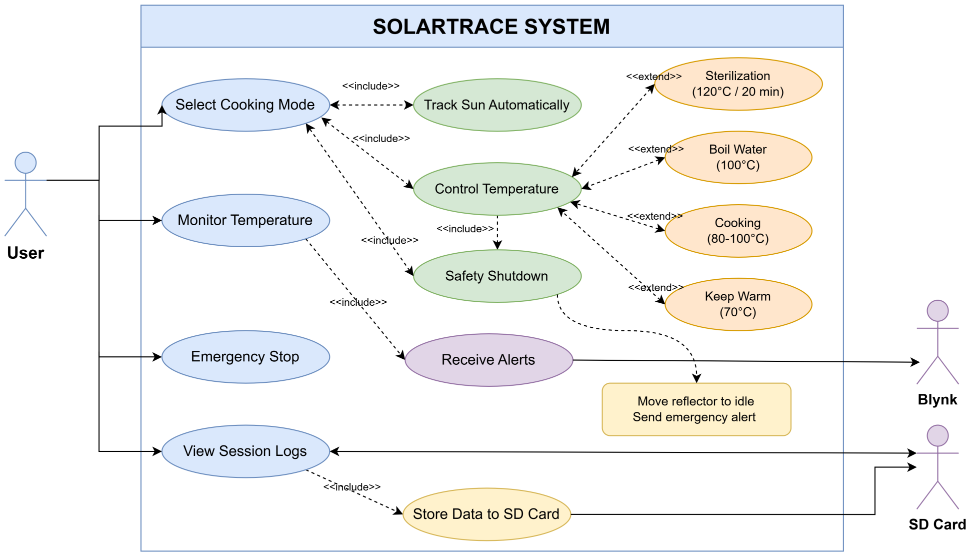

Figure 8: SolarTrace Use Case Diagram

7.3 Use Case Descriptions

Table 8: Use Case Descriptions

Use Case	Actor	Description
Select Mode	User	Choose between Cooking, Sterilization, Boil Water, or Warming modes
Monitor Temperature	User	View real-time temperature from thermocouple on mobile app
Receive Alerts	User	Get push notifications when target temperature is reached or safety triggered
View Logs	User	Access historical session data stored on SD card
Track Sun	System	Automatically position reflector based on RTC time
Log Data	System	Record session data including temperature and duration
Safety Mode	System	Move reflector to idle position when temperature exceeds 130°C

8 ACTIVITY DIAGRAMS

8.1 Cooking Cycle Activity Diagram

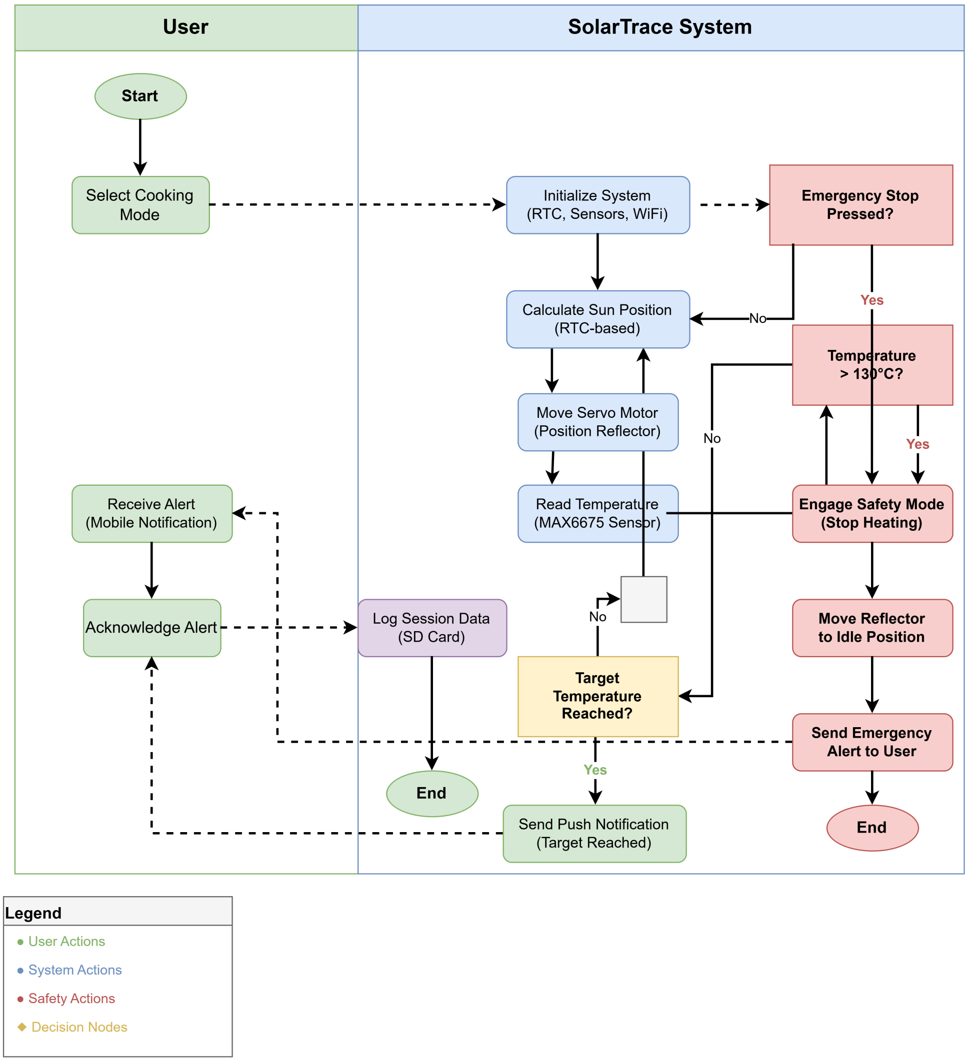

Figure 9: Activity Diagram: Cooking Cycle

8.2 Sterilization Mode Activity Diagram

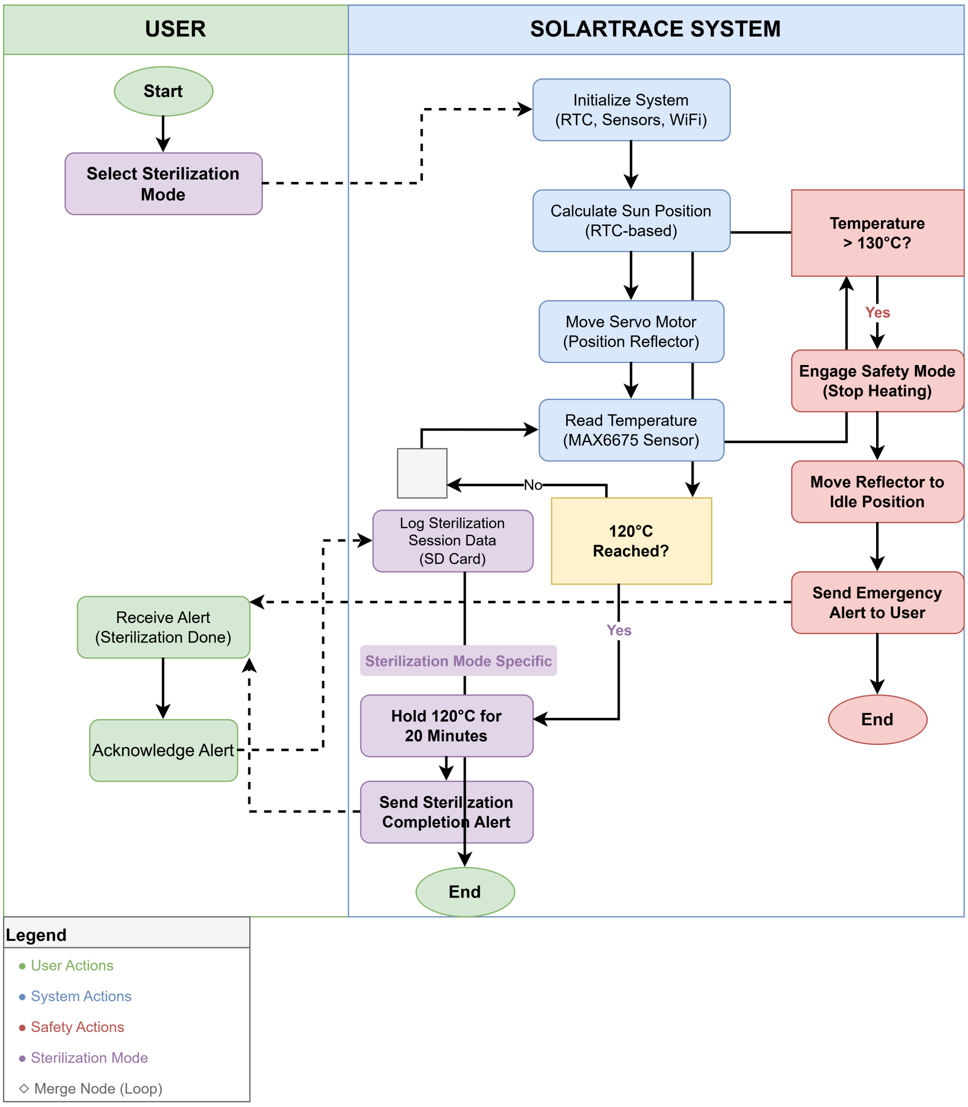

Figure 10: Activity Diagram: Sterilization Mode

9 SEQUENCE DIAGRAMS

9.1 Mode Selection Sequence

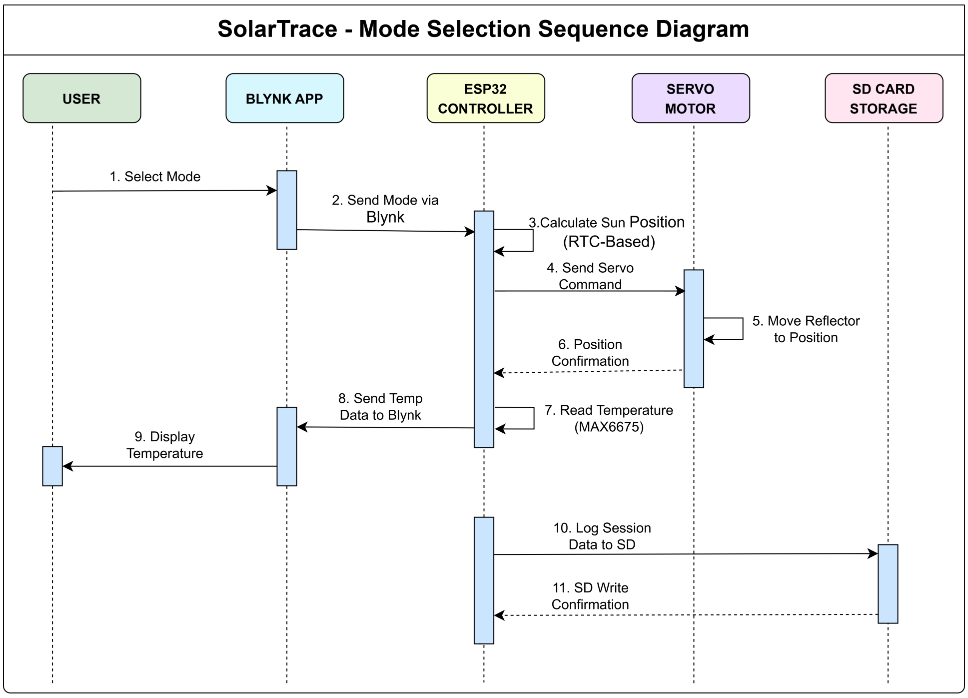

Figure 11: Sequence Diagram: Mode Selection

9.2 Temperature Alert Sequence

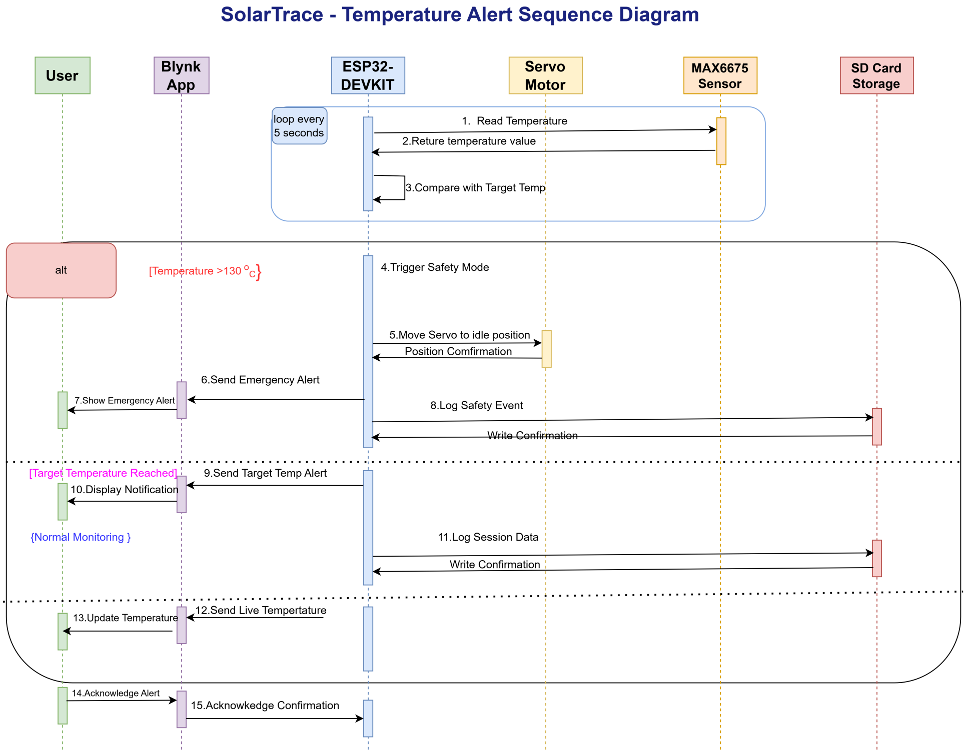

Figure 12: Sequence Diagram: Temperature Alert

10 CLASS DIAGRAM

10.1 Logical Class Structure

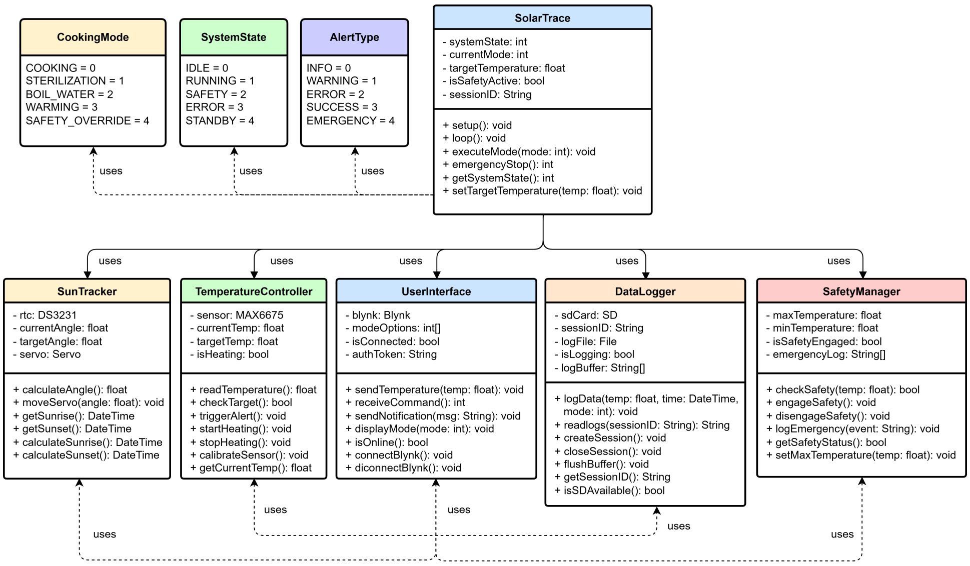

Figure 13: Class Diagram: Logical Structure

10.2 Class Descriptions

Table 9: Class Descriptions and Responsibilities

Class	Responsibilities
SolarTrace	Main controller; manages system state and mode execution
SunTracker	Handles RTC communication; calculates solar angle; controls servo motor
TemperatureController	Reads thermocouple data; checks target temperatures; triggers alerts
DataLogger	Manages SD card operations; logs session data; retrieves historical logs
UserInterface	Handles Blynk communication; sends temperature data; receives user commands

11 PACKAGE DIAGRAM

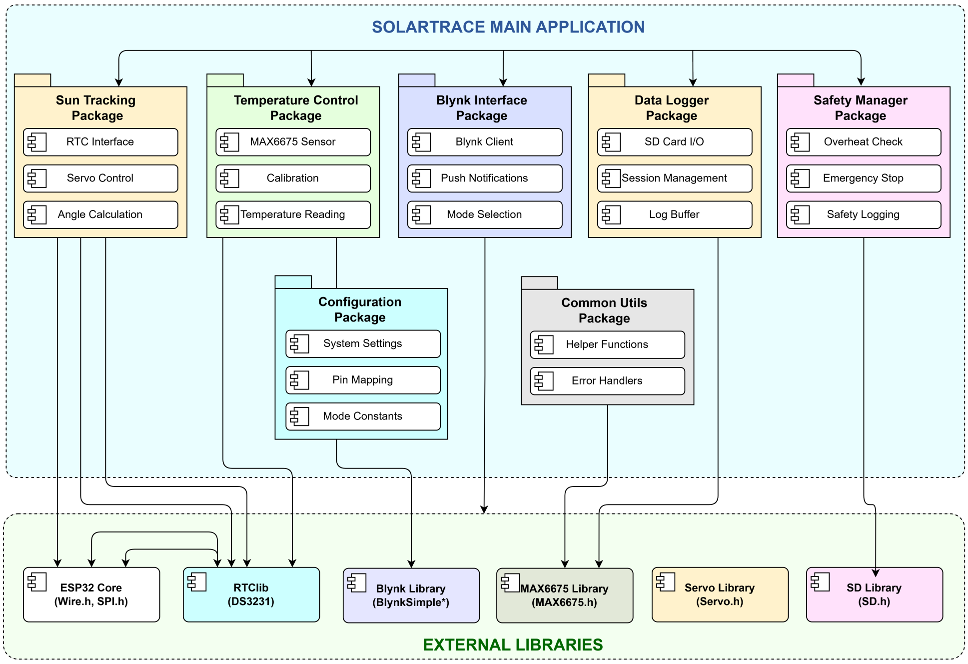

Figure 14: Package Diagram: Module Organization

11.1 Package Descriptions

Table 10: Package Descriptions

Package	Contents
Main Application	Main entry point; initialization; system loop; state management
Tracking	RTC communication; solar angle calculation; servo positioning
Sensors	MAX6675 interface; temperature conversion; error handling
Communication	Blynk integration; push notifications; mode selection; data streaming
Data Storage	SD card operations; session creation; data logging; log retrieval

12 TECHNICAL SPECIFICATIONS

12.1 Hardware Specifications

Table 11: Hardware Technical Specifications

Component	Model	Specifications
Microcontroller	ESP32-DEVKIT	Dual-core Xtensa LX6, 240MHz, 520KB SRAM, 4MB Flash, Wi-Fi/BLE
RTC Module	DS3231	±2ppm accuracy, I2C interface, battery backup, 0-70°C range
Temperature Sensor	MAX6675	12-bit resolution, 0.25°C accuracy, 0-500°C range, SPI interface
Thermocouple	K-Type	Chromel-Alumel, -200 to 1350°C range, 41 µV/°C sensitivity
Servo Motor	MG995	15kg/cm torque, 0.2s/60° speed, 4.8-7.2V, metal gears
Power System	5V Panel + 18650	5W panel, 3.7V 3000mAh battery, TP4056 charger
Data Storage	MicroSD Card	Up to 32GB, FAT32 format, SPI interface

12.2 Pin Configuration for ESP32-DEVKIT

Table 12: ESP32-DEVKIT Pin Configuration

ESP32-DEVKIT Pin	Function	Component	Notes
GPIO21	I2C Data (SDA)	DS3231 RTC	4.7k pull-up required
GPIO22	I2C Clock (SCL)	DS3231 RTC	4.7k pull-up required
GPIO13	PWM Output	MG995 Servo	50Hz signal, 0-180° control
GPIO15	SPI Chip Select	MAX6675	Active low
GPIO14	SPI Clock	MAX6675	
GPIO12	SPI Data Out	MAX6675	
GPIO5	SPI Chip Select	MicroSD	
GPIO18	SPI Clock	MicroSD	
GPIO23	SPI Master Out	MicroSD	
GPIO19	SPI Master In	MicroSD	
VIN (5V)	Power Input	MG995 Servo	5V power
3.3V	Power Output	RTC, MAX6675, SD	3.3V power
GND	Ground	All	Common ground

12.3 Tracking Algorithm Specification

Table 13: Tracking Algorithm Parameters

Parameter	Value
Latitude	4.85°N (Juba, South Sudan)
Sunrise Time	6:30 AM (local)
Sunset Time	6:00 PM (local)
Total Daylight	690 minutes (11.5 hours)
Sun Movement	15° per hour
Servo Adjustment Frequency	Every 10 minutes
Servo Step Angle	2.5° per step
Total Tracking Range	0° to 120°
Servo Update Formula	angle = (minutes_since_sunrise / total_daylight) × 120°

12.4 Cooking Mode Specifications

Table 14: Cooking Mode Parameters

Mode	Code	Target Temperature	Special Conditions
Cooking	0	80-100°C	General cooking of staple foods
Sterilization	1	120°C	Hold for 20 minutes; completion alert
Boil Water	2	100°C	Alert when reached; no hold required
Warming	3	70°C	Maintain temperature; keep warm
Safety Override	4	N/A	Move reflector to idle; 130°C trigger

12.5 Communication Protocol

Table 15: Blynk IoT Communication Parameters

Parameter	Value
Platform	Blynk IoT
Protocol	TCP/IP over Wi-Fi
Data Rate	Configurable (default: 1 second interval)
Virtual Pins	V0-Mode Selection, V1-Temperature, V2-Notifications, V3-Logging, V4-Safety
Authentication	Auth Token from Blynk App
Push Notifications	Blynk.notify() function
Offline Operation	SD card logs; sync when reconnected

13 INTERFACE DESIGN

13.1 Blynk Dashboard Layout

Table 16: Blynk Widget Configuration

Widget	Pin	Function
Dropdown	V0	Mode selection: Cooking, Sterilization, Boil Water, Warming
Gauge	V1	Real-time temperature display (0-150°C)
Notification	V2	Push notification alerts
Label	V3	Session logging status display
Button	V4	Emergency stop/override
Label	V5	CO savings estimate display

13.2 User Interaction Flow

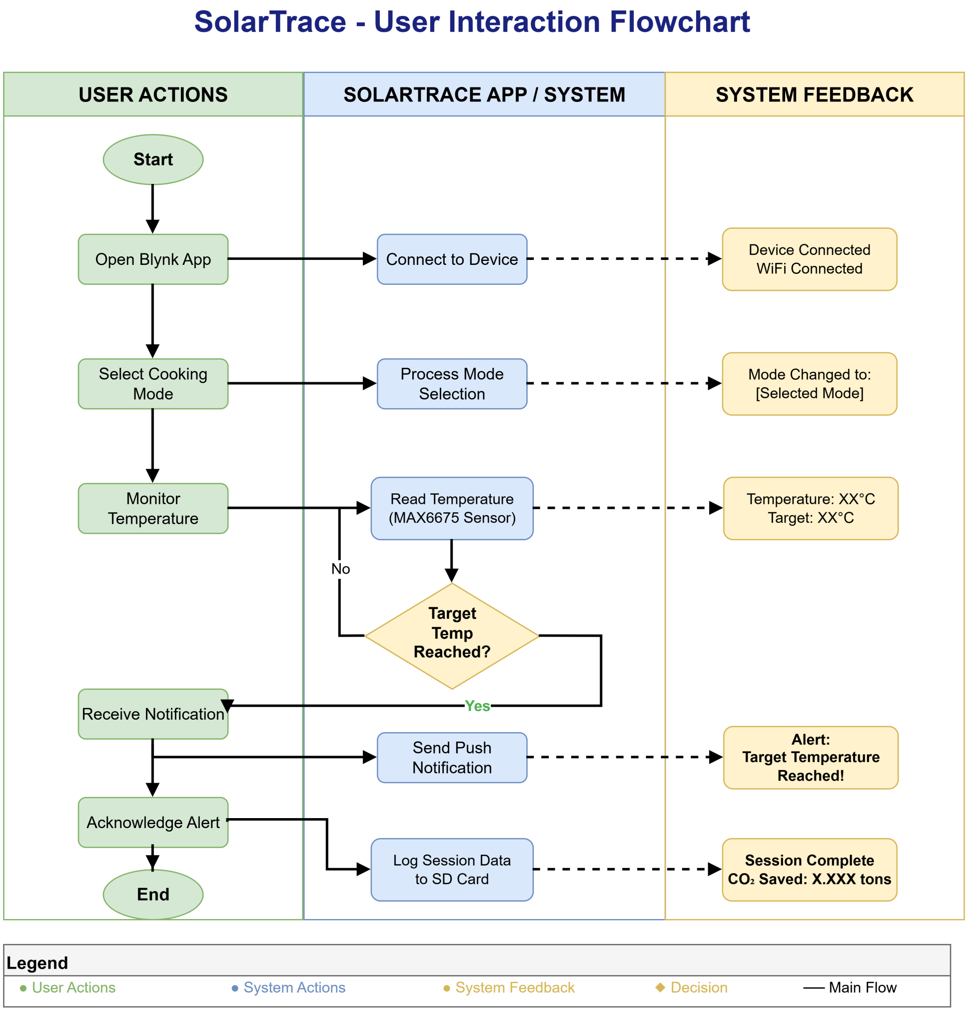

Figure 15: User Interaction Flow

14 SUMMARY AND NEXT STEPS

14.1 Design Summary

This chapter has presented the complete technical architecture and design for the SolarTrace system, including:

- **Alternative Systems Analysis**: Validating the SolarTrace approach against existing solutions
- **ESP32-DEVKIT Microcontroller**: Detailed pinout, specifications, and features
- **System Schematic**: Complete wiring diagram and component connections
- **System Architecture**: Three-tier architecture with physical, control, and application layers
- **UML Diagrams**: Context, data flow, use case, activity, sequence, class, and package diagrams
- **Technical Specifications**: Detailed component, pin, algorithm, and mode specifications
- **Interface Design**: Blynk-based mobile application with intuitive controls

14.2 ESP32-DEVKIT Advantages for SolarTrace

Table 17: ESP32-DEVKIT Advantages

Feature	Benefit for SolarTrace
Built-in WiFi	Enables IoT connectivity without additional modules
Built-in BLE	Supports Bluetooth for local configuration
Dual-core Processor	Handles tracking and communication simultaneously
Low Cost	Under $25, suitable for rural deployment
Large Community	Extensive documentation and libraries available
Low Power Consumption	Suitable for solar/battery operation

14.3 NEXT STEPS: Week 3 - Core Development

With the design phase complete, the team will proceed to Week 3: Core Development, which includes:

1. Setting up the development environment (Arduino IDE, libraries, Blynk)
2. Implementing RTC initialization and time retrieval

3. Coding servo positioning logic based on time-of-day calculations
4. Integrating temperature monitoring with MAX6675
5. Testing each component independently
6. Basic system integration

14.4 Design Decision Justification

Table 18: Key Design Decisions and Justifications

Decision	Justification
RTC-based tracking vs. Light sensors	More reliable in cloudy conditions; lower cost; predictable operation
ESP32-DEVKIT vs. Arduino Nano	Built-in Wi-Fi reduces external components; more processing power; development board ready
Blynk platform	Free tier suitable for prototype; easy to use; push notification support
K-type thermocouple	Wide temperature range; suitable for high-temperature cooking
SD card logging	Offline capability essential for areas with limited internet connectivity
MG995 servo	High torque sufficient for 0.9m reflector; metal gears for durability

15 REFERENCES

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Document Control

Version	Changes	Date	Author
1.0	Initial Week 2 submission package	July 31, 2026	Team South Sudan
1.1	Added ESP32 pinout, system schematic, wiring diagram	July 31, 2026	Team South Sudan
1.2	Updated to ESP32-DEVKIT, added specifications table	August 1, 2026	Team South Sudan

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This document represents the complete Week 2 submission package for the UbuntuNet Alliance Women Hackathon 2026.

TEAM SOUTH SUDAN - SOLARTRACE
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